

Sunny “Number Basher” Chang Lu

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EDUCATION

Shanghai Starriver Bilingual School

Shanghai, China

Class of 2027 | GPA: XXX (weighted) | ACT: 34 | TOEFL: 116

AP Research (2026–27); 11 of 12 AP exams scored 5.

Audited CIS 160 and CIS 121 at University of Pennsylvania; COS 126 at Princeton University.

Summer Programs: Ross '26 | ESPR '25 | PACT Year-Round '21, '22 | Yau Math Camp '25 | Yingcai Talent Program '25

HONORS & AWARDS

Silver Medal, Scientific Round, International Olympiad in Artificial Intelligence 2024

Sixth Term Examination Papers (STEP) II: 115/120 (S); STEP III: 102/120 (S) 2026

AIME: 12/15; AMC 12: 138/150 2026

Winner, 2nd International Obfuscated Python Code Contest 2024

Finalist, Best Pen & Paper Puzzle, Thinky Awards 2025

High Commendation, John Locke Institute Essay Competition 2024

1st Team Overall and 2nd Individual, Duke Math Meet 2025

RESEARCH

Kempe’s Universality Theorem on Restricted Topologies (not available online) 2025–2026

- Proved that linkages remain universal when restricted to crossing-free, bipartite graphs of maximum degree three.
- Built interactive Desmos constructions of each gadget.

HeXO is PSPACE-Hard 2025

- Proved $\text{Connect}(n, 6, 2, 1)$ on a hexagonal grid PSPACE-complete by reduction from Quantified 3SAT.

Folding Polyominoes into Uniform-Thickness Rectangles; Tiling Squares with Polyominoes 2025

ACTIVITIES & LEADERSHIP

Contributor, PowerPoint Vexillology (Arglin’s Cenote) 2025–Present

Contributed to a 2,000+ member research effort constructing every flag with PowerPoint tools; supplied the SmartArt cycle construction necessary for flags of Croatia and Kyrgyzstan; presented the work in a talk on the Math Club channel

Puzzle Designer 2023–Present

Origami Puzzles | Thinky Awards finalist | contributor to PUZZDLE | playtester for Herd and Workworkwork

Presenter of 2swap’s Math Club 2025–Present

Seminars on tilings, foldings, and PowerPoint constructibility; 100,000+ total views on YouTube

OEIS Contributor 2025–Present

Proposed sequence A398486; contributed corrections and comments to existing entries

Open-Source Developer 2023–Present

Origami Simulator, HeXO Renderer, and miscellaneous projects

Speaker, TEDxXuhui 2023

Participant, LIVE Communication Training 2025–Present

WORK EXPERIENCE

Worksheet Author, starmathcat (YouTube mathematics channel) 2026–Present

Mathematics Tutor, Cambridge STEP Exam Preparation (five sessions) May 2026

Freelance Translator, Checkmating Combinations in Chess into Chinese Aug–Dec 2024

SKILLS

Programming: Python, Desmos, $\text{\LaTeX}$, Kotlin, Fundamentals of C++, Java, JavaScript, HTML

Languages: English, Chinese (Mandarin), toki pona